

Supplementary Information for “Conditional Layer Invariance in Protein Language Models: Position Leakage and Readout Choice Determine the Apparent Benefit of Multi-Scale Embeddings”

S1. Literature anchors for the revised estimates

The revised estimates are anchored to published values as follows. Random-split masked-position Ridge probing of 650M PLMs on ProteinGym has been reported at $\rho \approx 0.60$ (this work, submitted version: 0.596–0.607; unified rerun: 0.603–0.606); our random-position lookup control reaches 0.600 under the same protocol, quantifying how much of that level is attributable to position identity. Validated zero-shot (logit-lens style) scoring with 650M models is typically reported at moderate positive correlations with DMS fitness; our corrected fp32 final-layer estimates (ESM-2 0.443, SaProt 0.437, $n = 63$) fall in that range, whereas the withdrawn fp16 ESM-2 curve (-0.021 at the final layer) did not. Layer-fusion claims of large gains (e.g., a 12% Spearman improvement for attention-based fusion in multi-chain interaction prediction) are task- and protocol-dependent; under leakage-free evaluation of mutation effects we find no strategy exceeding the last layer.

Table 1: Anchoring of revised numbers to published values.

Quantity	Published anchor	This work (revised)
Random-split probing level	$\rho \approx 0.60$ (650M PLMs)	0.603–0.606 (all arms)
Position lookup control	—	0.600 (random), 0.10–0.14 (leakage-free)
Zero-shot lens, final layer	moderate positive	ESM-2 0.443; SaProt 0.437
Fusion gain over last layer	up to 12% (other tasks)	$\leq +0.018$ (Ridge), never best
Structure tokens vs. masked	—	+0.06 to +0.12 (leakage-free only)

S2. Full 33-layer mean Spearman ρ (combined features)

Table 2: ESM-2 650M: per-layer mean ρ by arm and split (best layer per column in bold).

Layer	masked_wt Random	masked_wt Modulo	masked_wt Contig.	masked_wt Pos.-gr.	unmasked_wt Random
0	0.601	0.232	0.067	0.224	0.601
1	0.601	0.235	0.129	0.219	0.602
2	0.602	0.262	0.148	0.252	0.602
3	0.602	0.275	0.155	0.267	0.603
4	0.603	0.285	0.159	0.276	0.603
5	0.602	0.301	0.165	0.290	0.603
6	0.603	0.307	0.167	0.298	0.603
7	0.603	0.312	0.169	0.303	0.603
8	0.603	0.316	0.174	0.306	0.603
9	0.603	0.322	0.180	0.308	0.603
10	0.603	0.324	0.195	0.315	0.604
11	0.603	0.334	0.198	0.323	0.604
12	0.603	0.332	0.201	0.326	0.604
13	0.603	0.339	0.204	0.327	0.603
14	0.603	0.336	0.205	0.328	0.603
15	0.603	0.340	0.203	0.331	0.603
16	0.603	0.344	0.199	0.333	0.604
17	0.604	0.348	0.196	0.341	0.604
18	0.604	0.347	0.190	0.339	0.604
19	0.604	0.350	0.206	0.341	0.604
20	0.604	0.359	0.210	0.352	0.604
21	0.604	0.364	0.213	0.354	0.604
22	0.604	0.364	0.215	0.355	0.604
23	0.604	0.368	0.228	0.361	0.604
24	0.604	0.377	0.236	0.369	0.604
25	0.604	0.387	0.248	0.378	0.604
26	0.604	0.398	0.268	0.389	0.605
27	0.605	0.406	0.281	0.399	0.605
28	0.605	0.424	0.313	0.414	0.605
29	0.605	0.435	0.327	0.426	0.606
30	0.606	0.449	0.350	0.440	0.606
31	0.606	0.453	0.365	0.446	0.606
32	0.606	0.459	0.370	0.450	0.606

Table 3: SaProt (structure-masked): per-layer mean ρ by arm and split (best layer per column in bold).

Layer	masked_wt Random	masked_wt Modulo	masked_wt Contig.	masked_wt Pos.-gr.	unmasked_wt Random
0	0.600	0.241	0.048	0.229	0.601
1	0.602	0.230	0.137	0.221	0.602
2	0.602	0.250	0.151	0.246	0.602
3	0.602	0.262	0.166	0.255	0.602
4	0.602	0.285	0.176	0.275	0.602
5	0.602	0.292	0.172	0.285	0.603
6	0.602	0.307	0.174	0.300	0.603
7	0.603	0.317	0.178	0.311	0.603
8	0.603	0.335	0.197	0.327	0.603
9	0.603	0.339	0.200	0.332	0.603
10	0.603	0.342	0.193	0.335	0.603
11	0.603	0.352	0.205	0.344	0.603
12	0.602	0.358	0.212	0.350	0.603
13	0.602	0.359	0.219	0.353	0.603
14	0.602	0.364	0.229	0.360	0.603
15	0.602	0.366	0.233	0.362	0.603
16	0.602	0.370	0.234	0.364	0.603
17	0.602	0.371	0.240	0.367	0.603
18	0.603	0.374	0.247	0.368	0.603
19	0.602	0.374	0.250	0.368	0.603
20	0.602	0.374	0.249	0.370	0.603
21	0.602	0.374	0.244	0.370	0.603
22	0.602	0.377	0.249	0.373	0.603
23	0.602	0.381	0.253	0.377	0.603
24	0.602	0.379	0.254	0.376	0.603
25	0.602	0.378	0.252	0.374	0.603
26	0.602	0.383	0.256	0.379	0.603
27	0.603	0.387	0.261	0.384	0.604
28	0.603	0.389	0.266	0.383	0.604
29	0.603	0.388	0.271	0.383	0.604
30	0.603	0.391	0.277	0.383	0.604
31	0.603	0.391	0.282	0.385	0.604
32	0.603	0.386	0.274	0.382	0.604

Table 4: SaProt (real 3Di): per-layer mean ρ by arm and split (best layer per column in bold).

Layer	masked_wt Random	masked_wt Modulo	masked_wt Contig.	masked_wt Pos.-gr.	unmasked_wt Random
0	0.601	0.274	0.112	0.251	0.602
1	0.603	0.299	0.188	0.289	0.603
2	0.603	0.337	0.221	0.326	0.604
3	0.603	0.346	0.233	0.336	0.604
4	0.603	0.364	0.250	0.357	0.604
5	0.604	0.372	0.264	0.365	0.604
6	0.604	0.379	0.271	0.372	0.604
7	0.604	0.387	0.272	0.378	0.604
8	0.604	0.398	0.284	0.390	0.604
9	0.604	0.398	0.290	0.392	0.604
10	0.604	0.399	0.288	0.393	0.605
11	0.604	0.407	0.298	0.399	0.605
12	0.605	0.411	0.306	0.404	0.605
13	0.604	0.416	0.311	0.408	0.605
14	0.605	0.420	0.317	0.413	0.605
15	0.605	0.421	0.321	0.414	0.605
16	0.605	0.424	0.326	0.417	0.605
17	0.605	0.426	0.332	0.420	0.605
18	0.605	0.427	0.335	0.421	0.605
19	0.605	0.428	0.336	0.422	0.605
20	0.605	0.431	0.343	0.425	0.605
21	0.605	0.433	0.345	0.429	0.605
22	0.605	0.436	0.352	0.432	0.606
23	0.605	0.439	0.358	0.434	0.606
24	0.605	0.442	0.365	0.437	0.606
25	0.605	0.443	0.365	0.438	0.606
26	0.605	0.447	0.369	0.441	0.606
27	0.605	0.452	0.375	0.445	0.606
28	0.605	0.455	0.379	0.448	0.606
29	0.605	0.456	0.385	0.450	0.605
30	0.605	0.459	0.392	0.454	0.605
31	0.606	0.461	0.394	0.454	0.605
32	0.606	0.460	0.394	0.453	0.605

S3. Per-dataset results (mutant arm, modulo split)

Per-dataset best-layer and final-layer Spearman ρ for the mutant arm under the modulo split. Complete per-dataset results for every arm, model, split, and layer are archived at the Zenodo deposit and in the project repository.

Table 5: Per-dataset mutant-arm probing (modulo split, combined features). Best layer (and ρ) and final-layer (L32) ρ .

Dataset	n	ESM-2			SaProt (3Di)		
		best L	best ρ	L32 ρ	best L	best ρ	L32 ρ
A0A1I9GEU1_NEIME_Kennouche_2019	922	0	0.087	-0.077	3	0.004	-0.006
A0A2Z5U3Z0_9INFA_Doud_2016	5000	0	0.442	0.415	23	0.457	0.361
A4GRB6_PSEAI_Chen_2020	5000	32	0.548	0.548	31	0.751	0.677
A4_HUMAN_Seuma_2021	5000	15	0.674	0.623	3	0.677	0.656
AACC1_PSEAI_Dandage_2018	1801	29	0.470	0.426	32	0.549	0.549
ADRB2_HUMAN_Jones_2020	5000	31	0.507	0.506	31	0.527	0.526
AMIE_PSEAE_Wrenbeck_2017	5000	31	0.574	0.519	31	0.640	0.631
B3VI55_LIPST_Klesmith_2015	5000	31	0.562	0.477	32	0.592	0.592
BLAT_ECOLX_Deng_2012	4996	29	0.417	0.338	29	0.551	0.544
BLAT_ECOLX_Firnberg_2014	4783	31	0.656	0.600	30	0.799	0.768
BLAT_ECOLX_Jacquier_2013	989	31	0.581	0.559	29	0.739	0.729
BLAT_ECOLX_Stiffler_2015	4996	31	0.668	0.630	31	0.783	0.720
CALM1_HUMAN>Weile_2017	1813	30	0.154	0.132	6	0.141	0.126
CCDB_ECOLI_Adkar_2012	1176	30	0.613	0.580	30	0.694	0.639
CCDB_ECOLI_Tripathi_2016	1663	30	0.459	0.443	30	0.546	0.442
CP2C9_HUMAN_Amorosi_abundance_2021	5000	32	0.555	0.555	32	0.671	0.671
CP2C9_HUMAN_Amorosi_activity_2021	5000	31	0.598	0.583	29	0.631	0.601
DLG4_HUMAN_Faure_2021	5000	6	0.729	0.654	23	0.670	0.657
DLG4_RAT_McLaughlin_2012	1576	29	0.471	0.404	14	0.530	0.503
DYR_ECOLI_Thompson_plusLon_2019	2363	30	0.371	0.342	25	0.455	0.407
ESTA_BACSU_Nutschel_2020	2172	17	0.349	0.320	32	0.478	0.478
F7YBW8_MESOW_Aakre_2015	5000	12	0.299	0.238	19	0.255	0.231
GAL4_YEAST_Kitzman_2015	1195	30	0.471	0.451	32	0.448	0.448
GCN4_YEAST_Staller_induction_2018	5000	8	0.692	0.649	2	0.659	0.614
GFP_AEQVI_Sarkisyan_2016	5000	13	0.494	0.467	32	0.488	0.488
GRB2_HUMAN_Faure_2021	5000	8	0.656	0.591	32	0.629	0.629
HIS7_YEAST_Pokusaeva_2019	5000	12	0.629	0.605	32	0.649	0.649
HSP82_YEAST_Flynn_2019	5000	29	0.453	0.449	23	0.476	0.439
HSP82_YEAST_Mishra_2016	4323	27	0.507	0.465	31	0.501	0.486
IF1_ECOLI_Kelsic_2016	1367	31	0.334	0.316	17	0.485	0.245
KKA2_KLEPN_Melnikov_2014	4960	32	0.640	0.640	23	0.654	0.651
MK01_HUMAN_Brenan_2016	5000	0	0.517	0.360	10	0.353	0.342
MSH2_HUMAN_Jia_2020	5000	31	0.521	0.519	22	0.468	0.446
MTH3_HAEAE_Rockah-Shmuel_2015	1777	31	0.627	0.618	31	0.667	0.665
NUD15_HUMAN_Suiter_2020	2844	27	0.448	0.386	31	0.674	0.651
P53_HUMAN_Giacomelli_NULL_Etoposide_2018	5000	0	0.597	0.471	32	0.611	0.611
P53_HUMAN_Giacomelli_NULL_Nutlin_2018	5000	31	0.637	0.634	14	0.618	0.613
P53_HUMAN_Giacomelli_WT_Nutlin_2018	5000	31	0.812	0.810	28	0.830	0.826
P53_HUMAN_Kotler_2018	1048	31	0.684	0.594	30	0.693	0.675
P84126_THETH_Chan_2017	1519	32	0.497	0.497	31	0.457	0.448
PABP_YEAST_Melamed_2013	5000	10	0.673	0.637	32	0.675	0.675
PTEN_HUMAN_Matreyek_2021	5000	30	0.559	0.553	27	0.597	0.575
PTEN_HUMAN_Mighell_2018	5000	30	0.536	0.510	31	0.539	0.529
Q59976_STRSQ_Romero_2015	2999	30	0.622	0.597	32	0.665	0.665
RASH_HUMAN_Bandaru_2017	3134	19	0.494	0.448	26	0.583	0.524
RL401_YEAST_Mavor_2016	1253	13	0.520	0.448	22	0.601	0.513
RL401_YEAST_Roscoe_2013	1195	30	0.603	0.532	17	0.629	0.484
RL401_YEAST_Roscoe_2014	1380	0	0.382	0.317	24	0.532	0.448
SC6A4_HUMAN_Young_2021	5000	30	0.569	0.552	32	0.569	0.569
SPG1_STRSG_Olson_2014	5000	21	0.669	0.653	32	0.672	0.672
SRC_HUMAN_Ahler_CD_2019	3372	30	0.504	0.417	32	0.573	0.573
SUMO1_HUMAN>Weile_2017	1700	30	0.414	0.414	32	0.447	0.447
SYUA_HUMAN_Newberry_2020	2497	13	0.537	0.503	10	0.546	0.538
TADBP_HUMAN_Bolognesi_2019	1196	0	0.712	0.582	0	0.689	0.679
TPK1_HUMAN>Weile_2017	3181	32	0.251	0.251	19	0.220	0.199
TPMT_HUMAN_Matreyek_2018	3648	30	0.552	0.502	23	0.541	0.518
TPOR_HUMAN_Bridgford_S505N_2020	562	7	0.400	0.236	12	0.481	0.387
TRPC_SACS2_Chan_2017	1519	32	0.542	0.542	31	0.560	0.530
TRPC_THEMA_Chan_2017	1519	32	0.358	0.358	7	0.372	0.279
UBC9_HUMAN>Weile_2017	2563	30	0.392	0.360	31	0.451	0.448

S4. TOST equivalence tests (all leakage-free comparisons)

Two one-sided tests on paired per-dataset differences between the best layer (fixed a priori as the argmax of the layer-mean curve) and layer 32; margin $\Delta\rho = 0.01$; 90% bootstrap confidence intervals. No test shows a difference beyond the margin.

Table 6: TOST results; 7 of 19 comparisons are equivalent at the margin, 12 indeterminate.

Model	Arm	Split	Best	n	Mean Δ	90% CI	TOST p	Verdict
ESM-2	mutant	Modulo	L31	63	0.004	[-0.003, +0.011]	0.049	Equivalent
ESM-2	mutant	Contiguous	L31	57	0.006	[-0.006, +0.019]	0.25	Indeterminate
ESM-2	mutant	Pos.-grouped	L31	63	0.003	[-0.006, +0.012]	0.059	Indeterminate
SaProt (mask)	masked_wt	Modulo	L30	63	0.005	[+0.000, +0.011]	0.045	Equivalent
SaProt (mask)	masked_wt	Contiguous	L31	57	0.008	[+0.003, +0.013]	0.188	Indeterminate
SaProt (mask)	masked_wt	Pos.-grouped	L31	63	0.003	[-0.000, +0.007]	0.001	Equivalent
SaProt (mask)	unmasked_wt	Modulo	L30	63	0.006	[+0.000, +0.013]	0.136	Indeterminate
SaProt (mask)	unmasked_wt	Contiguous	L31	57	0.004	[-0.002, +0.011]	0.041	Equivalent
SaProt (mask)	unmasked_wt	Pos.-grouped	L27	63	0.003	[-0.005, +0.011]	0.052	Indeterminate
SaProt (mask)	mutant	Modulo	L30	63	0.005	[-0.003, +0.013]	0.098	Indeterminate
SaProt (mask)	mutant	Pos.-grouped	L30	63	0.004	[-0.006, +0.014]	0.11	Indeterminate
SaProt (3Di)	masked_wt	Modulo	L31	63	0.002	[-0.002, +0.005]	0.0	Equivalent
SaProt (3Di)	masked_wt	Contiguous	L31	57	0.001	[-0.005, +0.007]	0.001	Equivalent
SaProt (3Di)	masked_wt	Pos.-grouped	L31	63	0.002	[-0.002, +0.005]	0.0	Equivalent
SaProt (3Di)	unmasked_wt	Modulo	L31	63	0.007	[+0.003, +0.011]	0.081	Indeterminate
SaProt (3Di)	unmasked_wt	Contiguous	L31	57	0.009	[+0.003, +0.015]	0.328	Indeterminate
SaProt (3Di)	unmasked_wt	Pos.-grouped	L31	63	0.008	[+0.004, +0.013]	0.254	Indeterminate
SaProt (3Di)	mutant	Modulo	L31	63	0.006	[+0.000, +0.012]	0.101	Indeterminate
SaProt (3Di)	mutant	Contiguous	L31	57	0.006	[-0.005, +0.016]	0.218	Indeterminate
SaProt (3Di)	mutant	Pos.-grouped	L31	63	0.008	[+0.002, +0.014]	0.24	Indeterminate

S5. Foldseek 3Di token preparation disclosure

For each ProteinGym substitution dataset, the wild-type sequence was aligned to its AlphaFold Protein Structure Database entry and each residue was assigned the Foldseek 3Di state of its aligned structural residue. Structures were obtained for 62 of 63 datasets. Two alignments were truncated at the C terminus and missing residues were padded with the Foldseek gap state (UBC9: 1 residue; Q59976: 22 residues). The influenza nucleoprotein dataset has no AFDB entry and was evaluated with structure-masked tokens in all SaProt analyses.

Table 7: 3Di files: length and gap-padded residue count per dataset.

Dataset	3Di length	Gaps	Dataset	3Di length	G
A0A1I9GEU1_NEIME_Kennouche_2019	161	0	MSH2_HUMAN_Jia_2020	934	
A4GRB6_PSEAI_Chen_2020	266	0	MTH3_HAEAE_Rockah-Shmuel_2015	330	
A4_HUMAN_Seuma_2021	770	0	NUD15_HUMAN_Suiter_2020	164	
AACC1_PSEAI_Dandage_2018	177	0	P53_HUMAN_Giacomelli_NULL_Etoposide_2018	393	
ADRB2_HUMAN_Jones_2020	413	0	P53_HUMAN_Giacomelli_NULL_Nutlin_2018	393	
AMIE_PSEAE_Wrenbeck_2017	346	0	P53_HUMAN_Giacomelli_WT_Nutlin_2018	393	
B3VI55_LIPST_Klesmith_2015	439	0	P53_HUMAN_Kotler_2018	393	
BLAT_ECOLX_Deng_2012	286	0	P84I26_THETH_Chan_2017	254	
BLAT_ECOLX_Firnberg_2014	286	0	PABP_YEAST_Melamed_2013	577	
BLAT_ECOLX_Jacquier_2013	286	0	PTEN_HUMAN_Matreyek_2021	403	
BLAT_ECOLX_Stiffler_2015	286	0	PTEN_HUMAN_Mighell_2018	403	
CALM1_HUMAN>Weile_2017	149	0	Q59976_STRSQ_Romero_2015	501	
CCDB_ECOLI_Adkar_2012	101	0	RASH_HUMAN_Bandaru_2017	189	
CCDB_ECOLI_Tripathi_2016	101	0	RL401_YEAST_Mavor_2016	128	
CP2C9_HUMAN_Amorosi_abundance_2021	490	0	RL401_YEAST_Roscoe_2013	128	
CP2C9_HUMAN_Amorosi_activity_2021	490	0	RL401_YEAST_Roscoe_2014	128	
DLG4_HUMAN_Faure_2021	724	0	SC6A4_HUMAN_Young_2021	630	
DLG4_RAT_McLaughlin_2012	724	0	SPG1_STRSG_Olson_2014	448	
DYR_ECOLI_Thompson_plusLon_2019	159	0	SRC_HUMAN_Ahler_CD_2019	536	
ESTA_BACSU_Nutschel_2020	212	0	SUMO1_HUMAN>Weile_2017	101	
F7YBW8_MESOW_Aakre_2015	93	0	SYUA_HUMAN_Newberry_2020	140	
GAL4_YEAST_Kitzman_2015	881	0	TADBP_HUMAN_Bolognesi_2019	414	
GCN4_YEAST_Staller_induction_2018	281	0	TPK1_HUMAN>Weile_2017	243	
GFP_AEQVI_Sarkisyan_2016	238	0	TPMT_HUMAN_Matreyek_2018	245	
GRB2_HUMAN_Faure_2021	217	0	TPOR_HUMAN_Bridgford_S505N_2020	635	
HIS7_YEAST_Pokusaeva_2019	220	0	TRPC_SACS2_Chan_2017	248	
HSP82_YEAST_Flynn_2019	709	0	TRPC_THEMA_Chan_2017	252	
HSP82_YEAST_Mishra_2016	709	0	UBC9_HUMAN>Weile_2017	159	
IF1_ECOLI_Kelsic_2016	72	0	VKOR1_HUMAN_Chiasson_abundance_2020	163	
KKA2_KLEPN_Melnikov_2014	264	0	VKOR1_HUMAN_Chiasson_activity_2020	163	
MK01_HUMAN_Brenan_2016	360	0	YAP1_HUMAN_Araya_2012	504	